Supplementary Figure S5 | Splicing and cross-platform protein evidence define a mediation boundary

Discordant protein assays constrain — rather than invalidate — the RNA-centered locus model.

A Splicing and isoform audit

No validated soluble-IL12RB2 mechanism was identified

IL12RB2 sQTL

No rs6679356–IL12RB2 sQTL in GTEx v10; detected sQTLs belong to SERBP1.

No splice-mediated causal chain can be claimed.

Soluble transcript

Ensembl transcripts lacking the canonical transmembrane segment are annotation-level candidates only.

Retain only as a testable mechanism.

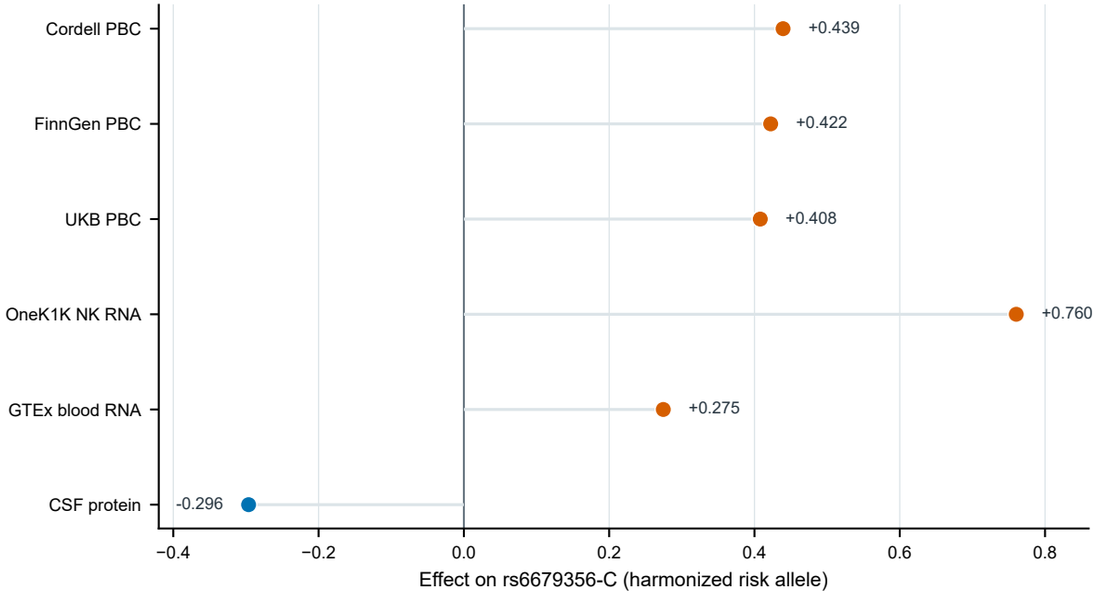
Reviewed UniProt isoform

Reviewed truncated isoforms retain a transmembrane segment and are not validated soluble receptors.

Do not label reviewed isoforms as soluble IL12RB2.

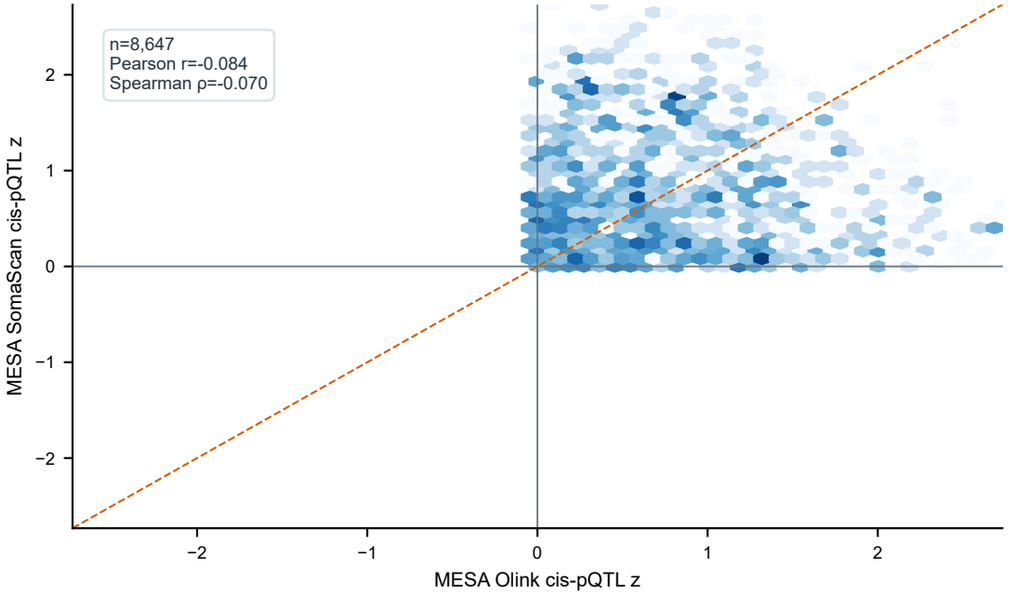
B Disease–RNA–protein direction chain

CSF protein is opposite to disease and RNA directions



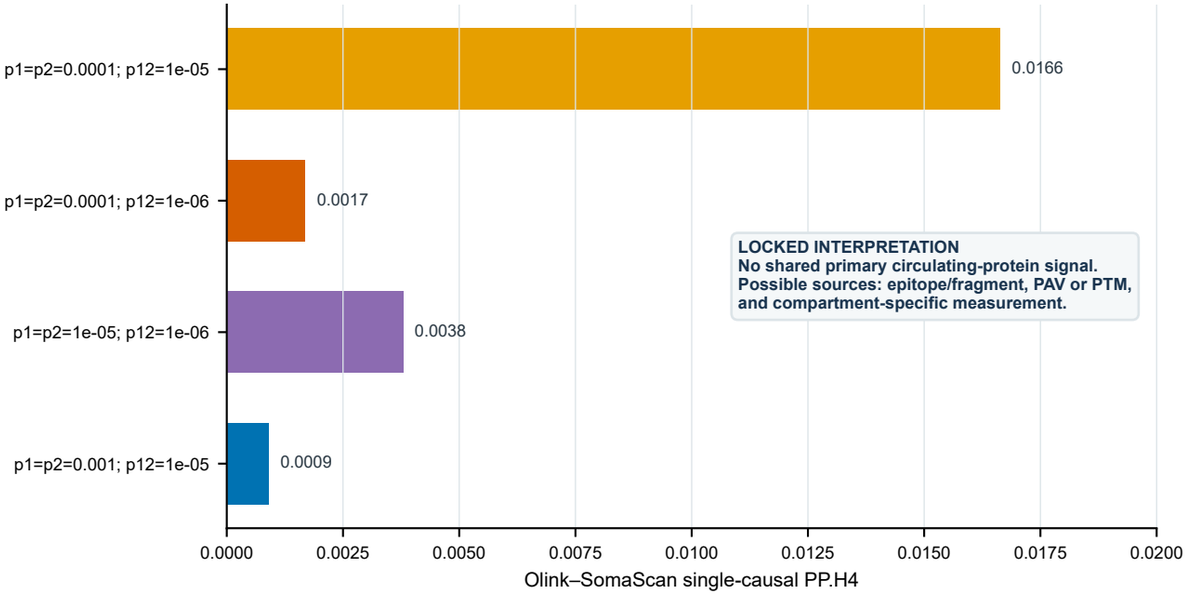
C Same-sample cross-platform comparison

8,647 aligned cis variants; density shown on a log scale



D Protein signal-level sensitivity

All tested priors remain non-supportive



Interpretation boundary: CSF or circulating SomaScan/Olink effects cannot be used to infer cell-surface IL12RB2 abundance or pharmacologic direction. These results reject a simple RNA→circulating protein→PBC mediation chain.